# Final Degree Project Documentation in Computer Engineering in the AI Era

An unofficial, opinionated and practical guide for TFG documentation at USAL

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1st Edition • 2026

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This English version is a structural placeholder. The first complete version of the book is maintained in Spanish.

**Note**

Translation pending. Always use the Spanish version as the source of truth for now.

The book is an unofficial, opinionated and practical guide for documenting a Final Degree Project in Computer Engineering at the University of Salamanca.

# Part I

# TFG Regulations

# TFG REGULATIONS

Translation pending. Use the Spanish version as the source of truth.

## 1.1 USAL Regulations

Translation pending. Use the Spanish version as the source of truth.

## 1.2 Faculty of Sciences Regulations

Translation pending. Use the Spanish version as the source of truth.

## 1.3 Department Regulations

Translation pending. Use the Spanish version as the source of truth.

## 1.4 Professional Standards

Translation pending. Use the Spanish version as the source of truth.

# Part II

# TFG Documentation

# TFG DOCUMENTATION

Translation pending. Use the Spanish version as the source of truth.

## 2.1 Style Rules

Translation pending. Use the Spanish version as the source of truth.

## 2.2 Citations and Bibliography

Translation pending. Use the Spanish version as the source of truth.

## 2.3 Current Structure

Translation pending. Use the Spanish version as the source of truth.

## 2.4 Old ITIS/PFC Structure

Translation pending. Use the Spanish version as the source of truth.

## 2.5 Old vs Current Mapping

Translation pending. Use the Spanish version as the source of truth.

CHAPTER 3

# THE REPORT

Translation pending. Use the Spanish version as the source of truth.

## 3.1 Introduction

Translation pending. Use the Spanish version as the source of truth.

## 3.2 Purpose and objectives

Translation pending. Use the Spanish version as the source of truth.

## 3.3 Background and current situation

Translation pending. Use the Spanish version as the source of truth.

## 3.4 Methods, tools, and models

Translation pending. Use the Spanish version as the source of truth.

## 3.5 Requirements, scope, and constraints

Translation pending. Use the Spanish version as the source of truth.

## 3.6 Proposed solution

Translation pending. Use the Spanish version as the source of truth.

## 3.7 Risks and management

Translation pending. Use the Spanish version as the source of truth.

## 3.8 Conclusions

Translation pending. Use the Spanish version as the source of truth.

# ANNEXES

Translation pending. Use the Spanish version as the source of truth.

## 4.1 A1. System specifications

Translation pending. Use the Spanish version as the source of truth.

## 4.2 A2. System analysis and design

Translation pending. Use the Spanish version as the source of truth.

## 4.3 A3. Size and effort estimation

Translation pending. Use the Spanish version as the source of truth.

## 4.4 A4. Security plan

Translation pending. Use the Spanish version as the source of truth.

## 4.5 A5. Other annexes

Translation pending. Use the Spanish version as the source of truth.

# Part III

# Software Engineering

# SOFTWARE ENGINEERING APPLIED TO THE TFG

Translation pending. Use the Spanish version as the source of truth.

## 5.1 RUP vs Agile

Translation pending. Use the Spanish version as the source of truth.

## 5.2 UML and Documentation as Code

Translation pending. Use the Spanish version as the source of truth.

## 5.3 Planning, Risks and Tests

Translation pending. Use the Spanish version as the source of truth.

# Part IV

# Tools

# DOCUMENTATION TOOLS

Translation pending. Use the Spanish version as the source of truth.

## 6.1 Team Word

Translation pending. Use the Spanish version as the source of truth.

## 6.2 Team LaTeX

Translation pending. Use the Spanish version as the source of truth.

### 6.2.1 Starting templates in Overleaf

Public Overleaf templates can be useful as technical starting points for folder structure, LaTeX configuration, bibliography handling, and PDF layout [Overleaf26b]. They are not official USAL or DIA templates. When using an external template, adapt it to the current TFG rules, review its license, remove institutional text that does not apply, and cite the original template.

Examples worth reviewing include the University of Seville TFG template [Tri26], the ULL-ESIT-GII TFG report template [Tor26], the University of Málaga TFG report template [Hid26], the University of Cádiz software-development TFG template [ProfesoradodDdIIdlUdCadiz26], and the UDIMA Computer Engineering TFG template [Bar26].

### 6.2.2 Overleaf from VS Code

If you want to write in VS Code while keeping the project in Overleaf, the cleanest route is usually Overleaf's Git integration, which lets you clone the project, work locally, and synchronize changes with `pull` and `push` [Overleaf26a]. Another option is the community extension Overleaf Workshop, which opens Overleaf/ShareLaTeX projects in VS Code and supports editing, compiling, previewing, and collaboration features [iam26].

Treat any extension-based workflow as an external tool: check authentication, permissions, project state, and conflicts before relying on it for final TFG work.

## 6.3 VS Code, AI Agents and Skills

Translation pending. Use the Spanish version as the source of truth.

# Part V

# Mistakes and FAQ

# COMMON MISTAKES AND FAQ

Translation pending. Use the Spanish version as the source of truth.

## 7.1 Common Mistakes

Translation pending. Use the Spanish version as the source of truth.

## 7.2 FAQ

Translation pending. Use the Spanish version as the source of truth.

# Part VI

# Defense

# TFG PRESENTATION AND DEFENSE

Translation pending. Use the Spanish version as the source of truth.

# Part VII

# Final Information

# CREDITS

Author: **Alvaro Lozano Murciego**.

Publication year: **2026**.

This English section is a placeholder. The Spanish version is the source of truth for the first public version of the book.

# LICENSES

# HOW TO CITE

Suggested citation:

> Lozano Murciego, A. (2026). *Documentacion de un TFG en Ingenieria Informatica en tiempos de IA*. Universidad de Salamanca. CC BY 4.0.

```
@book{lozano2026tfgia,
  author = {Lozano Murciego, Alvaro},
  title = {Documentacion de un TFG en Ingenieria Informatica en tiempos de IA},
  year = {2026},
  publisher = {Universidad de Salamanca},
  note = {Unofficial guide, CC BY 4.0}
}
```

# BIBLIOGRAPHY


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